Shreyansh Samaddar

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LinkedIn | GitHub | Portfolio

Education

International Institute of Information Technology — Naya Raipur, CG, India
B.Tech in Data Science and Artificial Intelligence — *Aug 2024 – Present*

- Current CGPA: 9/10

Technical Skills

- **Programming Languages:** Python, C++, SQL, JavaScript
- **AI / ML:** Deep Learning, Reinforcement Learning, Computer Vision, Generative AI, Diffusion Models, RAG, LLM Fine-Tuning, Agentic AI
- **Frameworks & Libraries:** PyTorch, TensorFlow, Scikit-Learn, Gymnasium, Hugging Face, Pandas, NumPy
- **Tools & Platforms:** Git, Linux, Docker, FastAPI, Apache Superset

Experience

- **Cybersecurity AI/ML Research Intern** — Mar 2026 – May 2026
  SwiftSafe, Hyderabad, India
  - Worked on an AI-agent orchestration platform (RLM layer) spanning the full RAG → LLM → Tooling pipeline for automated cybersecurity investigation.
  - Built dynamic tool selection, execution management, and result aggregation workflows enabling the LLM to perform multi-step autonomous reasoning across security-analysis modules.
  - Integrated capabilities including browser automation, network analysis, reverse engineering, OCR, sandbox execution, malware analysis, cloud auditing, and remote system investigation.
  - Collaborated with RAG (data ingestion, knowledge-base preparation) and LLM (fine-tuning, evaluation, inference) teams to coordinate end-to-end agentic pipeline integration.
- **Business Analysis & Data Analytics Intern** | Internship Certificate — Jun 2025 – Jul 2025
  Piramal Swasthya Management and Research Institute
  - Integrated a RAG-based AI system using Python, FastAPI, Apache Superset, and vector retrieval techniques to enable natural-language interaction for automated dashboard generation.
  - Implemented backend and API logic to convert user queries into structured dashboard configurations, reducing manual report-creation effort.
  - Contributed to full-stack development, supporting feature integration and platform stability.

Publications

Intelligent Cybersecurity SSH Honeypot with ML-Based Attack Recognition | *ARES 2026*

- Accepted at ARES 2026 — 21st International Conference on Availability, Reliability and Security (SecIndustry 2026 Track)
- Designed an SSH honeypot capable of realistic terminal simulation to capture and analyze attacker behavior at session level.
- Built a telemetry pipeline collecting commands, session metadata, and behavioral patterns from adversarial interactions.
- Developed ML-based attack classification models for automated threat categorization and behavioral analysis across captured sessions.
- Evaluated recurring intrusion techniques and attacker strategies from captured interaction logs.

Research & Projects

Mask-Guided Diffusion for Plant Disease Progression | *Vision, Generative AI* Jan 2026 – Feb 2026

- Developed a conversational framework for simulating plant disease progression using diffusion models with environment conditioning.
- Implemented progressive mask irregularization and fungicide intervention modeling for realistic disease-spread simulation.
- Achieved FID = 43.2, LPIPS = 0.14, lesion IoU = 94.7%, with zero healthy tissue modification across evaluated samples.

Reinforcement Learning Agent for TrackMania Nations Forever | *Deep RL* Dec 2025 – Feb 2026

- Developed a custom Gymnasium environment for TMNF using live telemetry to enable closed-loop reinforcement learning control.
- Designed a checkpoint-based reward structure penalizing crashes and incentivizing fast, smooth track completion.
- Trained deep RL agents (DQN) with checkpointing for long-running experiments; engineered compact state and action representations for stable autonomous driving behavior.

Lattice Gas Method Fluid Simulator | *Computational Physics* Dec 2025 – Jan 2026

- Developed a cellular automata–based fluid dynamics simulator using the Lattice Gas Method on discrete lattices.
- Implemented collision and streaming operators (FHP/HPP models) ensuring mass and momentum conservation.
- Validated macroscopic behavior against Navier–Stokes equations and benchmark CFD results.

Patents

Monitoring Air Pollution through Quantum Secured Encryption in IoT and Blockchain

- Indian Patent Granted | Patent No: 589285 | Date of Grant: 13 May 2026 | Co-Inventor
- Developed a blockchain-integrated architecture for tamper-resistant real-time environmental monitoring using quantum-secured encryption.

Achievements

Competitive Programming

- Codeforces Pupil rating

Extracurricular Activities

The Society of Coders IIIT-NR
Core Member *2025 – Present*

- Organized hackathons, coding competitions, and technical workshops.

Training and Placement Cell IIIT-NR
Junior Placement Coordinator *2026 – Present*

- Coordinating with companies and students to facilitate internship and placement processes.